Commuting at Pomona: A Dashboard for Sustainable Transportation Options

Data Section Draft

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1 Introduction

This section will introduce the problem statement and goals of the Sustainability Office [1] as described by the 2030 SAVE goals established. Some details will include metrics such as faculty and staff commuting making up 23% of Pomona's transportation emissions (3% faculty, 20% staff), and the college having 2030 goals of a 30% reduction in SOV commuting miles and a 100% reduction in net commuting emissions.

The goals of the project are a dashboard that helps employees see sustainable commute options from their place of residence, especially since this helps the sustainability office see where reductions are possible. In the introduction, I will also introduce the methodology of the dashboard and the various sources I scrape the public transit data from since there are various entities in Southern California.

2 Background

Additional context will be necessary including Pomona's 2026 Sustainability Annual Report [2], and the existing incentives (free Foothill Transit with the Class Pass, up to $100/month Metrolink reimbursement, and $2/day for alternative modes) which exist to encourage pulblic transit use, as well as the commute survey the data comes from. I will introduce the GTFS feeds and OpenTripPlanner that will be used in the project and define technical terms like SOV miles, etc so that the average reader can understand the more specific terms used in the project.

3 Data Draft

The data in this report comes from Pomona College's Sustainability Office, which surveyed faculty and staff in FY26 to quantify their commuting patterns and habits. The survey was done in line with The South Coast Air Quality Management District (AQMD) Rule 2202 [3], which states that institutions with 250 or more employees within the South Coast Air Basin must measure and reduce air pollution from employee commutes. Since vehicle emissions are a primary source of smog in Southern California, this data is collected to comply with Rule 2202, which encourages large employers to reduce the amount of single-occupancy vehicles heading to their worksites.

The data is in the form of two Excel workbooks, with one being for faculty and the other for staff. There is one sheet per fiscal year, originating

in FY14. For the purposes of this analysis the FY26 sheet is the main sheet used in this analysis. The main variables used for analysis are the ZIP code where the respondents report residing, their one-way miles to work, employee type, as well as a mode code for each weekday (Monday-Friday). FY14 will be used as the baseline to measure the progress that Pomona College has made towards its goal of a 30

Each respondent reported a letter code for each day of the week they commute to Pomona, such as H if they drive alone or J if they carpool with two people. For clarity, the data was reshaped so that each row represents one person on one day. Respondents that noted their commute on a given day as either driving alone or a solo motorcycle trip are noted as having taken a SOV trip, but zero-emission vehicles and carpools are excluded from this count. Days which included no commute such as a vacation or sick day were excluded from the count.

Faculty and staff combined together result in 467 total respondents, meaning 32 percent or 112 of 355 faculty completed the survey, and approximately 43 percent or 355 of 820 staff responded to the survey. To preserve anonymity, employee IDs are dropped and replaced with an anonymous person id. General data cleaning practices were conducted such as eliminating trailing spaces and truncating ZIP codes.

ZIP code boundaries come from the U.S. Census Bureau's 2020 ZIP Code Tabulation Areas, downloaded with the tigris R package [4, 5]. The map in the dashboard shows where respondents live and the share of their commute days spent driving alone in each ZIP code. To protect the privacy of respondents, ZIP codes with fewer than 10 respondents are shown in gray and percentages are rounded to the nearest 10 percent. In addition, respondents who live more than 40 miles from campus are excluded from the map. ZIP code 91711 (Claremont) has the most respondents and the lowest share of respondents who drive-alone, measuring at about 30 percent. Nearby faculty and staff who live in communities Upland, La Verne, and Pomona report around 70 percent drive-alone days despite these areas being only a few miles from campus. As such, these communities are the prime focus area to focus the dashboard around for transit and carpool alternative.

The dashboard includes progress made towards Pomona's 2030 goal of reducing SOV commuting since 2014 by 30 percent. This was done by comparing the FY14 baseline survey to FY26. Because not every employee took the survey, each year's survey results were scaled up to estimate the total for all employees in a full year following the Sustainability's Office method of multiplying the survey miles by the ratio of total employees to respondents, and then by the number of weeks worked per year which is 49

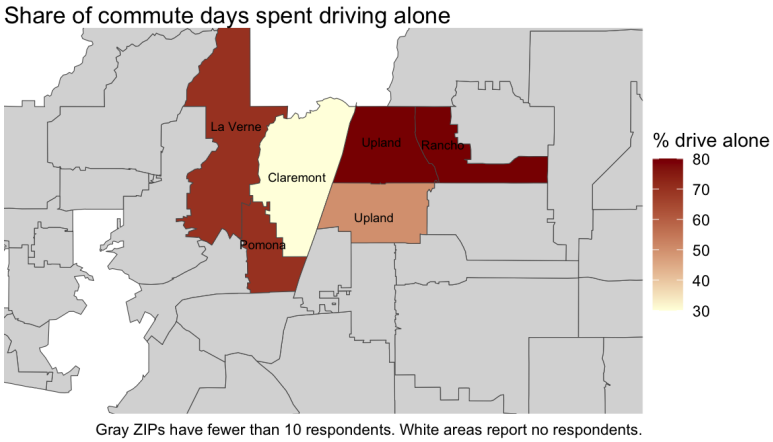


Figure 1: Share of commute days driving alone by ZIP code

for staff and 32 for faculty. Total annual SOV miles fell from about 1.78 million in FY14 to 1.63 million in FY26, which is an 8 percent decrease. Since Pomona has 1,175 employees now compared to 1,012 in 2014, SOV miles per employees fell by 21 percent. To reach the goal by 2030, total annual SOV miles need to fall to about 1.24 million, which would mean cutting approximately 390,000 SOV miles annually by in the next three years.

4 The Dashboard

The dashboard will be a public tool for Pomona faculty and staff to use in order to show realistic alternatives to driving alone from where they live. Since SOV miles are down only 8 percent since 2014, the goal of this tool is to encourage users to learn more about commuting alternatives that exist which they can implement and the incentives that exist to encourage use of the alternatives.

The user will see a map of home areas showing how people in the each area commute, colored by the share of commute days spent driving alone. Users will be able to click and select their area, and see a summary of how their coworkers there get to campus, along with the alternatives available from the chosen area. For transit, the dashboard will show the approximate travel time to campus by bus or train, or note that no practical transit option exists. For carpooling, it will show how many coworkers live in the

same area, which is only shown for areas more than about 3 miles from campus. Walking and biking options will only appear for areas close enough to campus for them to be realistic. Employees living 40 or more miles away will be grouped together and shown carpool and Metrolink options instead of a map area.

The incentives that apply for their modality will also be shown so users know what support is available for them. These include free Foothill Transit rides with the Class Pass, up to 100 dollars per month reimbursement for Metrolink, and 2 dollars per day for biking, walking, carpooling, taking transit, or driving a vehicle producing zero-emissions. Showing incentives next to each option connects the alternatives to a concrete benefits, which may further encourage less SOV rides.

This is built with R Shiny and a Leaflet map because Shiny turns R code into an interactive website, so the same code used to clean and analyze the survey can be used to power the dashboard. Leaflet adds a zoomable map with streets and town names, allowing users to find where they live and click on their area. The dashboard will use the FY26 survey data, Census ZIP code boundaries, and GTFS transit schedules from Foothill Transit and Metrolink, with transit travel times calculated using the r5r package.

5 Ethical Considerations

The use of this data set presents several ethical challenges to the locational privacy of data shared by faculty and staff at Pomona, requiring sensitive handling to prevent employee identification by their ZIP code. This is most specific to the 70 ZIP codes that each have only one person as reported by either a faculty or staff member. To prevent any ZIP codes identifying individual staff member locations, my dashboard only shows areas with 10+ respondents and groups the employees who live 40+ miles away into one group which is not mapped. Although this reduces the utility of the dashboard, protecting privacy is worth this tradeoff, since respondents never agreed to have their locations revealed publicly in a dashboard.

Furthermore, during the data cleaning process, employee IDs, names, and emails were stripped and replaced with an anonymous person ID so that survey data could not be traced to an individual's identity. Maintaining the understanding that employees filled out this survey to meet a regulation mandated by their employer means that the responses must be treated with the purpose of solely incentivizing faculty and staff to use sustainable means as their commute to work.

In addition, my dashboard acknowledges that living closer to Claremont, where there may be more sustainable transit options, is a privilege, since housing prices are higher near the College. My goal with this dashboard is to not blame individuals with few alternatives besides driving and instead show options and incentives which take aim to educate and nudge faculty and staff to reconsider the frequency in SOV use. Similarly, the dashboard is also transparent when an employee's area has no practical transit option.

6 Exploratory Analysis

This section shows the main commuting patterns. Faculty mostly live close to campus and already walk or bike often, while staff live farther away and mostly drive alone. I will show shares by distance and compare faculty and staff. I will also show that most SOV miles come from long staff commutes, which shapes where the 30% goal can realistically come from. There will be more analysis in this section as more is learned.

7 Transit and Alternatives Analysis

This section looks at what alternatives exist from each area with nearby transit stops and areas where several employees live close together and could carpool. These results are what the dashboard shows.

Limitations

- Commute distances are self-reported one-way miles.
- The survey covers one week, which may not be typical.
- Only 32 percent of faculty and 43 percent of staff responded.
- Scaling up assumes non-respondents commute like respondents.
- ZIP codes only give a rough location.

8 Conclusion

This section summarizes the findings and what they mean for the sustainability office, along with the limitations that exist which includes a small survey sample and no commute start times which can impact travel time due to traffic. I will end with suggestions for future work, like adding questions

to the survey about whether employees know about the incentives and how likely staff would be to reconsider their commuting patterns.

References

[1] Pomona College Sustainability Office. Faculty and Staff Commute Master Sheets, FY14–FY26. Unpublished data, 2026.

[2] Pomona College Sustainability Office. *2026 Sustainability Annual Report: Transportation*. 2026. https://tableau.campus.pomona.edu/views/2026-SAR/Transportation

[3] South Coast Air Quality Management District. *Rule 2202: On-Road Motor Vehicle Mitigation Options*. https://www.aqmd.gov/docs/default-source/rule-book/reg-xxii/rule-2202.pdf

[4] U.S. Census Bureau. 2020 ZIP Code Tabulation Areas, Cartographic Boundary Files.

[5] Walker, K. *tigris: Load Census TIGER/Line Shapefiles*. R package. https://CRAN.R-project.org/package=tigris

[6] Wickham, H., et al. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, 4(43), 1686.

[7] Wickham, H. and Bryan, J. *readxl: Read Excel Files*. R package.